

Lab Task– 2

Reinforcement Learning Laboratory

Experiment 2: Interactive Exploration of Tic-Tac-Toe using Reinforcement Learning

Duration: 3 Hours

Course Outcome (CO1): Identify the agent, environment, state, action, and reward in a Reinforcement Learning environment, and evaluate the effect of training episodes on the agent's learning performance.

RL Demo: <https://jinglescode.github.io/reinforcement-learning-tic-tac-toe/>

Task 1: Identify the Reinforcement Learning Components

Open the RL Tic-Tac-Toe demo and identify the following Reinforcement Learning components.

RL Component	Student Observation
Agent	
Environment	
State Representation	
Action Space	
Reward Function	
Learning Approach Used	

Answer the following questions:

1. Who is the learning agent in this environment?
2. What is considered as the environment?
3. How is the game state represented?
4. What are the possible actions available to the agent?
5. When does the agent receive a positive reward?
6. Which Reinforcement Learning approach is used for learning?
7. Is the agent trained using labelled data? Justify your answer.
8. Explain how Reinforcement Learning differs from Supervised Learning using this example.

Expected Observation

The RL agent plays Tic-Tac-Toe, the game board acts as the environment, the state is the current board configuration, actions correspond to placing a mark in an empty cell, and the agent learns through repeated self-play using rewards rather than labelled data.

Task 2: Effect of Training Episodes on Agent Behaviour

Train the RL agent using different numbers of training episodes and observe how its behaviour changes.

Training Episodes	Win %	Loss %	Draw %	Behaviour Observed
100				
500				
1,000				
5,000				
10,000				

Observe whether the agent:

- Plays randomly
- Blocks opponent moves
- Chooses the center position
- Chooses corner positions
- Makes strategic moves
- Avoids losing
- Forces a draw
- Plays optimally

Answer the following questions:

9. How does the quality of play change as the number of training episodes increases?
10. At approximately how many training episodes does the agent begin making intelligent decisions?
11. At what stage does the learning appear to converge?
12. Why does the win percentage improve with additional training?
13. Why do most games end in a draw after sufficient training?
14. What happens if the number of training episodes is too small?

Expected Observation

Initially, the agent plays randomly and loses frequently. As training increases, it learns to block the opponent, prefers the center and corners, avoids mistakes, and gradually converges to near-optimal play where most games end in a draw.

Submission Requirements

- Screenshots of the RL Tic-Tac-Toe demo
- Completed RL Components table
- Completed Training Episodes observation table
- Answers to all analysis questions
- A brief conclusion (5–6 lines) describing how the agent improves through self-play